

The recurrent ReNU syndrome allele is mutationally disfavoured and exclusively maternal in origin

Draft 2, 9 October 2026. Every number traces to a committed script, named in brackets. Not pre-registered; this is a separate study from the MuST-VEP variant-effect work and is reported as such.

Suggested venue. Short report / research letter: *Genetics in Medicine*, *European Journal of Human Genetics*, or *Human Molecular Genetics*. The result directly addresses a question posed in a 2026 *Nature* paper, so a Matters Arising there is also defensible.

Abstract

De novo variants in *RNU4-2* cause ReNU syndrome, among the most frequent single-gene causes of neurodevelopmental disorder yet described. A single-base insertion, n.64_65insT, accounts for 70–77% of all reported cases. The reason is unresolved: saturation genome editing has shown the variant is not unusually damaging, and the explanations named in the primary literature are an elevated local mutation rate, positive selection in the female germline, and ascertainment bias.

We quantified the mutational supply of this allele from public data. Base-pair-resolution germline mutation rates (Roulette) show the 18-nucleotide critical region is *less* mutable than the remainder of *RNU4-2* (mean relative rate 0.063 versus 0.088), and do not predict which critical-region variants are observed in patients or in population cohorts. Because n.64_65insT is a thymine inserted into a four-thymine homopolymer, we measured the single-base insertion frequency at homopolymer tracts empirically from gnomAD v4.1, across 10 genomic windows containing 10,992 T₄ tracts. Insertion frequency rises log-linearly with tract length (1.90-fold per added base, $R^2 = 0.90$, $\rho = 0.99$), reproducing the expected signature of replication slippage and validating the measurement. At tract length four, however, single-base insertions occur at only 0.40 times the frequency of substitutions — measured in both short-read gnomAD (0.23) and the long-read HPRC pangenome (0.40), of which we adopt the conservative long-read figure. Such sites are mutationally *disfavoured*.

The excess proves to be specific to a single allele rather than to insertions as a class: excluding n.64_65insT, the per-allele insertion-to-substitution ratio across the critical region is exactly 1.00. That permits a position-matched test, which holds assay coverage, calling behaviour and ascertainment constant by construction. Four variants are observed at chr12:120,291,839, and n.64_65insT accounts for **89 of the 93 patient carriers** there. Tested against the substitution at the same nucleotide — the comparison to which the measured mutational supply applies — the recurrent allele takes 89 of 91 carriers (97.8%, 95% CI 92.3–99.7%) where 28.7% is expected: exact binomial $p = 1.3 \times 10^{-45}$. The implied residual depends on how widely the substitution comparator is drawn, and we report the range rather than the most favourable value: **15-fold** using every substitution in the critical region, rising to 110-fold at the focal nucleotide alone, with $p \leq 2 \times 10^{-33}$ throughout. The two other insertions at that same nucleotide, which are the same class of variant call in the same homopolymer, carry two patients between them, excluding insertion-calling bias as the explanation. We further exclude interlocus gene conversion, the standard account of a recurrent allele with insufficient mutational supply: the tandem paralogue *RNU4-1* lies 1.2 kb away and is 97.2% identical, but carries the *identical* T₄ tract and differs from *RNU4-2* at no position inside the critical region, so conversion can generate none of the eleven observed patient alleles. *RNU4-1* also supplies a locus-specific calibration unavailable at most loci: it is callable, is not a disease gene, and at its homologous tract carries the exact equivalent insertion alongside two substitutions — one insertion per two substitutions, consistent with the genome-wide 0.40 used here, and evidence that the allele is readily generated at this context.

What remains is an allele-specific germline excess that is, in every informative case reported, **maternal**. The only documented mechanism capable of producing a germline excess of this magnitude — selfish spermatogonial selection — is paternal and depends on clonal expansion across hundreds of spermatogonial divisions accumulated over decades. Oogonial proliferation is confined to

fetal development and to roughly thirty divisions, after which the oocyte pool arrests, so an equivalent process would have to act far faster, within a bounded prenatal window, and reproducibly across unrelated women. No documented example of germline selection operates this way. Made quantitative, a selective account would require a per-division advantage of 17% across the roughly thirty prenatal oogonial divisions, against about 1.1% per division for the strongest documented selfish spermatogonial variants — fifteen times larger, in a window that closes before birth. We set out the three measurements that would identify the mechanism.

Introduction

RNU4-2 encodes the U4 small nuclear RNA, a component of the major spliceosome. De novo variants in an 18-nucleotide critical region cause ReNU syndrome, which accounts for roughly 0.4–0.5% of undiagnosed developmental disorder in large cohorts.

What is unusual is the allelic spectrum. One variant, n.64_65insT (GRCh38 chr12:120,291,839:T:TA), accounts for 77.4% of cases in the discovery cohort and 72.6% in an independent French cohort. Most Mendelian disorders are allelically heterogeneous; a single allele at this frequency demands an explanation.

The obvious one has been excluded by the primary literature. Saturation genome editing of *RNU4-2* found that n.64_65insT is not unusually damaging, with many variants scoring similarly or lower on function. Those authors concluded that high recurrence is unlikely to result from a particularly damaging functional effect driving ascertainment, and that positive selection in the female germline or an increased local mutation rate are the more likely explanations — while stating that which of these applies is at present unknown.

Here we quantify the mutational supply of the allele, because that is the term that can be settled with public data, and show that once it is measured the remaining discrepancy has a property that constrains its explanation severely.

Results

The critical region is not mutationally hot

We retrieved base-pair-resolution germline mutation rate estimates for every possible single-nucleotide variant in *RNU4-2* from Roulette, by range query against the per-chromosome rate file [48]. Across 423 records, the mean relative rate inside the 18-nucleotide critical region was 0.063, against 0.088 across the remainder of the gene. The region in which pathogenic variants cluster is therefore, if anything, *less* mutable than its surroundings.

Within the critical region, mutation rate did not predict which variants are observed. Rate was uncorrelated with the number of patient carriers (Spearman $\rho = +0.18$, $p = 0.41$, $n = 23$ variants) and with the number of population carriers ($\rho = -0.23$, $p = 0.29$). Observed variants separated cleanly by context rather than by rate: every variant with patient carriers had zero population carriers, and every variant with population carriers had zero patient carriers.

Insertion frequency at homopolymers scales log-linearly with tract length

Roulette models substitutions only. Since n.64_65insT is a thymine inserted into a T₄ homopolymer — the canonical signature of replication slippage — we measured the insertion frequency at homopolymer tracts directly [49]. Across 10 windows of chromosome 12 (~2 Mb of sequence), we enumerated every homopolymer tract from the reference and counted gnomAD v4.1 PASS single-base insertions assigned to each, expressing the result relative to the frequency of an observed substitution per possible substitution in the same windows — the same cohort, the same filters, the same ascertainment.

Insertion frequency rose monotonically and log-linearly with tract length ($\rho = 0.99$, $p = 2.6 \times 10^{-18}$; 1.90-fold per added base, $R^2 = 0.90$), from 0.10 at length two (A 0.094, T 0.113) to 24.5 at length twelve. This reproduces the established length dependence of replication slippage and serves as an internal validation of the measurement.

Table 1. Single-base insertion frequency at A/T homopolymer tracts, relative to the per-substitution frequency.

tract length	relative insertion frequency	tracts
2	0.09-0.11	149,527
3	0.11-0.13	64,305
4	0.21-0.26 (short read) / 0.26-0.54 (long read)	21,831
5	0.48-0.60	8,722
6	3.44-3.67	2,285
7	7.22-7.39	1,005
8	11.3-13.3	360

A T_4 tract is mutationally disfavoured, not favoured

At tract length four — the context of n.64_65insT — short-read data gave 0.257 times the substitution frequency for T_4 (163 insertions across 10,992 tracts; 95% Poisson CI 0.22–0.30) and 0.208 for A_4 (130 across 10,839; CI 0.17–0.24).

Because short-read pipelines are known to under-call insertions in homopolymers, we repeated the identical measurement in the same windows against the HPRC pangenome, whose variants derive from long-read haplotype assemblies [50]. Long-read data gave 0.543 for T_4 and 0.262 for A_4 , a mean of 0.403 — 1.73-fold above the short-read estimate. The ratio of long-read to short-read rate grew with tract length (1.3–2.2-fold at lengths 2–4, rising to 5.9–7.4-fold at lengths 10–12), the expected signature of short-read homopolymer failure, and is modest in the length range that matters here. We therefore adopt 0.40 as the expected relative insertion frequency at a T_4 tract. Insertions at such sites remain roughly 2.5-fold *less* likely than substitutions.

Interlocus gene conversion from the tandem paralogue is excluded

When a recurrent allele cannot be explained by mutation rate, the standard alternative is non-allelic gene conversion: a near-identical donor elsewhere in the genome repeatedly overwrites the acceptor, so the "mutation" is a copying event and its frequency is set by recombination rather than by polymerase error. This mechanism accounts for recurrent pathogenic alleles in *CYP21A2* (from *CYP21A1P*), *PMS2* (from *PMS2CL*), *SMN1* (from *SMN2*) and *GBA* (from *GBAP1*).

RNU4-2 has precisely the configuration that makes this plausible. Its paralogue *RNU4-1* lies 1,194 bp away on the same strand of chromosome 12, is the same length (141 nt), and is 97.2% identical (137/141).

The hypothesis is nonetheless decisively excluded, because conversion can only transfer sequence the donor actually carries [56]. *RNU4-1* differs from *RNU4-2* at exactly four positions — n.37, n.88, n.99 and n.113 — and **none lies inside the critical region**. In particular *RNU4-1* carries the *identical* four-thymine tract, so conversion reproduces T_4 rather than extending it to T_5 and cannot generate n.64_65insT. Applying the same test to every observed patient allele, **0 of 11 alleles and 0 of 114 patient carriers are conversion-explicable**.

The mutational supply is confirmed at a sequence-identical paralogous locus

A genome-wide average over 21,831 tracts need not apply to any one site: single-base insertion rates vary with flanking sequence, replication timing and chromatin. *RNU4-2* permits an unusually direct check on this. *RNU4-1* is sequence-identical across the critical region, lies 1.2 kb away, is callable in short-read data (2,545 gnomAD PASS records across 141 nt, unlike the *RNU1* family), and is not a disease gene — so its population variation is not depleted by selection against a phenotype. It

therefore reports the mutational supply at the same sequence context, at the same locus, free of disease ascertainment.

At the homologous T_4 tract (chr12:120,293,170–120,293,173), gnomAD contains **the exact equivalent single-base insertion** (chr12:120,293,173 T>TA, AC 5) together with two substitutions (both at chr12:120,293,170, AC 1 each) [60]. Counting distinct observed alleles — the unit used by the homopolymer analysis above, and the one least distorted by drift and allele age — the tract gives one insertion against two substitutions, a ratio of **0.50 against the genome-wide estimate of 0.40** applied here.

Two things follow. First, the insertion is demonstrably generated at this exact sequence context, so its absence from gnomAD at *RNU4-2* (AC 0; one UK Biobank carrier) reflects selection against the phenotype rather than an inability to arise. Second, the objection that this particular tract might be unusually insertion-prone is not supported by measurement at the one locus able to test it.

The caveat belongs alongside the result. By allele *count* rather than distinct alleles the tract gives 5 insertion alleles against 2 substitution alleles, because one insertion lineage has drifted upward while the substitutions are singletons. Allele count reflects drift and allele age as much as mutation rate, which is why distinct-allele counting is primary here, but a reader should see both numbers. Across the whole gene body the two paralogues agree closely: 0.091 insertions per substitution at *RNU4-2* and 0.064 at *RNU4-1*.

The excess is specific to one allele, and survives position-matched controls

Across the critical region, single-base insertions have 19.8 patient carriers per allele (5 alleles, 99 carriers) against 2.5 for pathogenic substitutions (6 alleles, 15 carriers) — a 7.9-fold excess per allele. That class-level comparison is, however, entirely attributable to one allele: **excluding n.64_65insT, the per-allele insertion-to-substitution ratio is exactly 1.00** (2.5 versus 2.5) [58]. The excess is not a property of insertions. It is a property of a single allele.

This permits a far stronger design than a class comparison. Because the effect is allele-specific, it can be controlled at the level of the **nucleotide**: variants at the same position are covered by the same assays, called by the same pipelines, reported in the same screens and ascertained together, so any explanation acting on a region, a variant class or a sequencing difficulty is held constant by construction.

chr12:120,291,839 provides exactly this control set. Four variants are observed there:

variant at chr12:120,291,839	class	patient carriers	population carriers
n.64_65insT	insertion extending the T_4 tract	89	1
n.64_65insG	insertion at the same nucleotide	2	0
n.64_65insC	insertion at the same nucleotide	0	2
n.65A>G	substitution at the same nucleotide	2	0

n.64_65insT accounts for **89 of the 93 patient carriers at this position**. The mutational supply measured above — a T_4 insertion occurring at 0.40 times the frequency of a substitution — speaks directly to the contrast between the recurrent insertion and the substitution at the same nucleotide, and there the recurrent allele takes **89 of 91 carriers** (97.8%, 95% CI 92.3–99.7%) where 28.7% is expected. The departure is extreme: exact binomial $p = 1.3 \times 10^{-45}$; by Poisson, 26.1 carriers are expected and 89 observed ($P(X \geq 89) = 5.2 \times 10^{-22}$).

The implied residual, however, depends on how widely the substitution comparator is drawn, and that dependence belongs in the paper rather than in a rebuttal. Widening it from the focal nucleotide to the whole critical region moves the residual from 110-fold to 15-fold:

substitution comparator	substitution carriers	observed share	residual	p
same nucleotide	2	0.978	110×	1.3×10^{-45}
the T ₄ tract	5	0.947	44×	6.3×10^{-42}
tract $\pm$ 3 nt	7	0.927	32×	7.0×10^{-40}
whole critical region	15	0.856	15×	1.9×10^{-33}

The narrowest comparator gives the largest number and should not be quoted alone. **We therefore take the conservative 15-fold as the headline residual**, note that it rises to 110-fold under position matching, and record that significance is overwhelming at every choice [58].

The two other insertions at this nucleotide are the decisive control. n.64_65insG and n.64_65insC are the same class of variant call, in the same homopolymer, in the same patients' data, and they carry 2 and 0 patients between them. Insertion-calling behaviour in this tract therefore cannot generate the observed excess; nor can assay coverage, since all four alleles occupy one base.

The excess is exclusively maternal

The parental origin of this excess is reported in the primary literature and is, so far as we are aware, without precedent in its consistency. In the discovery cohort, parental origin was resolved for 54 individuals — 46 carrying n.64_65insT, three other insertions, five substitutions — and **all 54 variants lay on the maternal allele**. In the independent French genome-sequencing cohort, origin was determined in 50 trios and one mother-patient duo: 47 maternal and four paternal, and **none of the four paternal variants was n.64_65insT** (they were n.62T>C, n.68A>C, n.76del and n.92C>G). The excess quantified above is therefore not merely associated with maternal transmission; within the resolution of the published data it occurs *only* on the maternal allele.

Discussion

The dominance of n.64_65insT in ReNU syndrome is not a consequence of an elevated mutation rate. The critical region is less mutable than its surroundings; mutation rate does not predict which variants are seen; and the specific sequence context of the recurrent allele is one in which insertions are disfavoured relative to substitutions by roughly 2.5-fold. Nor is it a consequence of interlocus gene conversion, the usual explanation for a recurrent allele with insufficient mutational supply: the only plausible donor carries the same tract. Two of the candidate explanations can therefore be set aside.

Ascertainment is the more mundane of those remaining and is the reason the analysis is position-matched. n.64_65insT was the allele through which the disorder was discovered and subsequent studies searched for it specifically, which inflates its apparent share; a comparison across the gene, or across variant classes, cannot separate that from a real excess. Restricting the comparison to a single nucleotide does. The three other variants at chr12:120,291,839 are reported in the same screens, called from the same reads by the same pipelines, and covered by any assay that covers the recurrent allele, yet carry four patients between them against 89. An ascertainment account must therefore explain why the bias acts on one allele while sparing two other insertions at the same base in the same homopolymer — including n.64_65insG, which differs only in the identity of the inserted nucleotide. Two further observations point the same way: the allele's share is similar in an independently ascertained cohort (72.6% versus 77.4%), and a single-base insertion in a homopolymer is the harder call for short-read pipelines, not the easier one, so the technical bias runs against the observed direction.

That leaves germline selection, and this is where the maternal exclusivity becomes the central problem.

The primary literature proposes positive selection in the female germline but does not quantify it. Our estimate puts a number on what it would have to achieve: a 15- to 110-fold enrichment of this allele over the substitutions against which it can be compared, beyond the mutational supply we measured. That magnitude is not unprecedented in itself. Selfish spermatogonial selection — the process

underlying the classical paternal age-effect disorders, driven by activating variants in *FGFR2* (Apert), *FGFR3* (achondroplasia), *RET* (MEN2) and other RAS-MAPK genes — produces enrichments of one to three orders of magnitude in offspring.

But that mechanism is a consequence of male germline architecture, and the female germline does not share it. Spermatogonial stem cells divide continuously throughout adult life — several hundred divisions by middle age — so a variant conferring even a slight proliferative advantage is amplified clonally across decades; this is why mutant cells form discrete patches in aged testis and why the effect scales with paternal age. Oogenesis offers a far smaller target. Oogonia do proliferate mitotically, so a proliferative compartment is not absent altogether, but that proliferation is confined to roughly the first half of fetal development and comprises on the order of thirty divisions, after which the oocyte pool enters meiotic arrest and the genome is not replicated again. A selective process would therefore have to achieve, within a bounded prenatal window, an enrichment that the male germline accumulates over hundreds of divisions and several decades — and would have to do so reproducibly across unrelated women. No documented example of germline selection operates this way.

The maternal age effect that does exist does not fill the gap. It is substantially weaker than the paternal one and is attributed to DNA damage accumulating in arrested oocytes and to double-strand-break-associated mutation clusters — processes that raise mutation rate broadly rather than amplifying one allele over its immediate neighbours. The position-matched comparison above bounds that directly: three other variants at the same nucleotide, subject to any such process equally, carry four patients between them.

One alternative deserves explicit treatment. The authors of the French cohort suggest the maternal bias may reflect negative selection against severely splicing-disruptive variants in the male germline. That would explain why ReNU variants are maternal; it does not explain why *one* maternal allele is 44-fold more frequent than the other maternal alleles at the same nucleotide, which would be purged equally. Directional selection against paternal transmission changes the parent of origin, not the allelic spectrum within a parent.

That asymmetry can be made quantitative rather than rhetorical [58, 60], and the requirement holds across the whole range of residuals above. Spread over the roughly thirty mitotic divisions separating a primordial germ cell from the arrested oocyte pool, a 15-fold enrichment needs a per-division advantage of **9.4%** and a 110-fold enrichment needs **17%**. The best-documented selfish spermatogonial variants reach up to a thousand-fold enrichment, but across approximately 610 spermatogonial divisions by paternal age forty — a per-division advantage of about **1.1%**. Even at our most conservative residual the female germline would have to sustain an advantage roughly **eight times larger per division** than the strongest documented male example, rising to fifteen times under position matching, and to do so entirely within a window that closes before birth. That the conclusion does not turn on which comparator is chosen is the point: this is the specific quantity any selective account must supply, and it is why we regard the explanation as open rather than merely unquantified.

We therefore state the problem rather than resolve it: ReNU syndrome exhibits a maternal-only germline excess of at least 15-fold, and up to 110-fold under position matching, specific to a single allele, at a site whose mutational supply we have measured and found to be below that of a substitution, and that survives controls at the resolution of the individual nucleotide. No described mechanism of germline mutation or selection produces that pattern.

Three measurements would discriminate among the possibilities. **(1) Maternal age.** If the excess arises from damage during meiotic arrest, the n.64_65insT fraction among ReNU cases should rise with maternal age; if it arises from a replication-independent process fixed before birth, it should not. Trio cohorts already hold the data. **(2) Direct assay of the female germline.** Deep targeted sequencing of *RNU4-2* in ovarian tissue or oocytes, as was done for *FGFR2* and *FGFR3* in testis, would show whether the allele is present above expectation before fertilisation — distinguishing a germline process from a post-zygotic or transmission one. **(3) Transmission distortion.** If the allele is favoured at fertilisation or in early embryogenesis rather than generated more often, that is

detectable in preimplantation or early-loss material, and would place the mechanism outside mutation entirely.

Limitations

Observed allele counts in population data reflect mutation rate, genetic drift and selection together, so our measurement is of relative mutability, not a per-generation rate. Short-read sequencing detects insertions in homopolymers less sensitively than substitutions; we measured that bias directly rather than assuming it, finding a 1.73-fold under-detection at length four, and adopted the long-read-corrected rate. The HPRC estimate rests on 232 assemblies, so its counts are sparse, and graph-based calls in repetitive sequence carry their own uncertainty; the true rate most likely lies between the two estimates, and we have taken the one that minimises our reported effect. Sampling was restricted to chromosome 12, and genome-wide sampling would tighten the estimate. Patient carrier counts were transcribed from a published table and inherit its ascertainment.

The parental-origin data are not ours: they are reported in the two cohort papers and we have combined them with our own measurement rather than re-deriving them. Origin was inferred indirectly from nearby informative variants, and was resolvable in only a subset of cases, so the claim is that no informative counter-example has been reported — not that none exists. The gene-conversion test excludes the proximal tandem donor; more distant U4 pseudogenes were not examined, though homology and proximity both make them far less plausible donors. Finally, we exclude mechanisms; we do not establish what the operative one is.

Figures

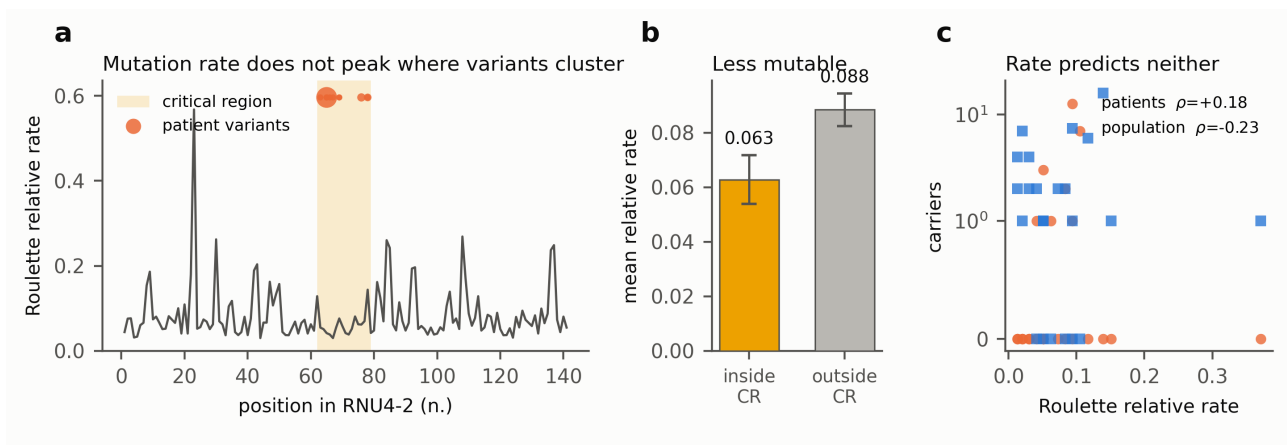


Figure 1. Mutation rate across *RNU4-2*. (a) Roulette relative rate per nucleotide with the 18-nt critical region shaded; patient variants overlaid. (b) Rate inside versus outside the critical region. (c) Rate versus patient carriers and versus population carriers, both flat.

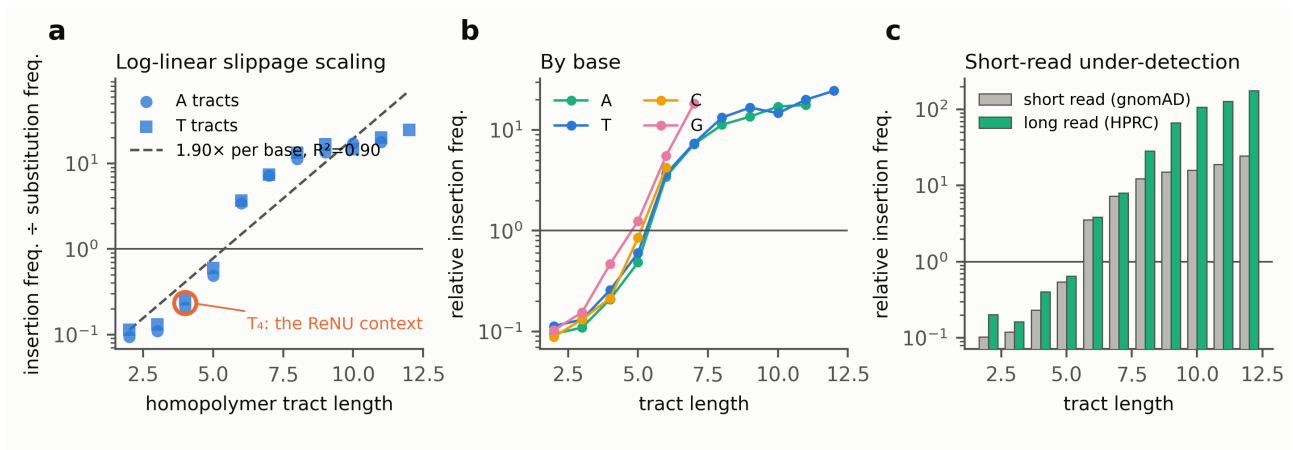


Figure 2. Homopolymer insertion frequency. (a) Relative insertion frequency against tract length, log scale, with the fitted slope; T_4 marked. (b) The same split by base. (c) Short-read versus long-read estimate at each tract length, showing the measured under-detection.

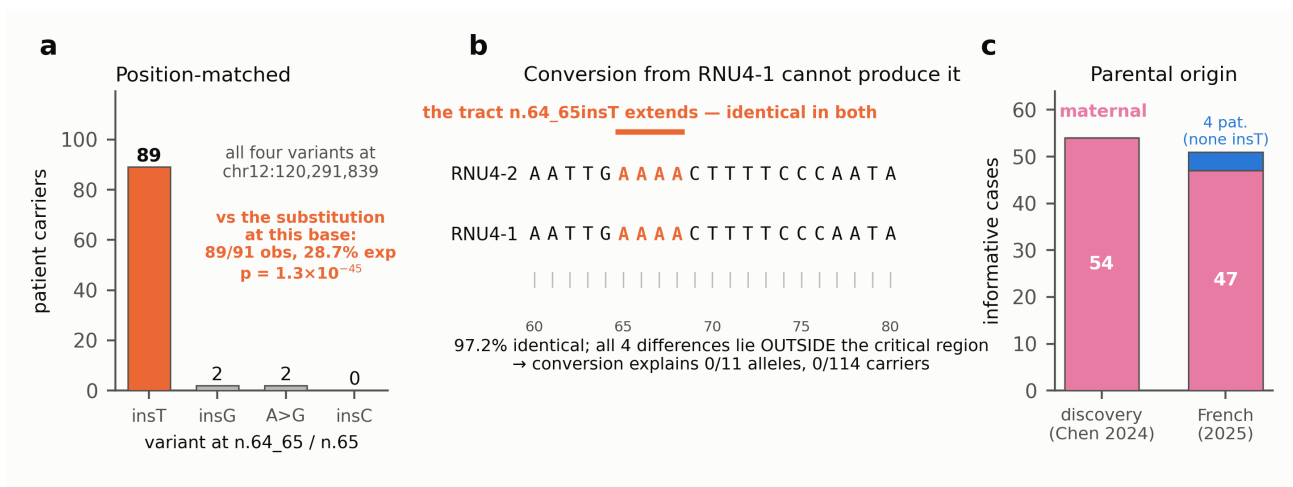


Figure 3. The position-matched residual and its parental origin. (a) All four variants observed at chr12:120,291,839, with the binomial test of the recurrent insertion against the substitution at the same nucleotide; the two other insertions at that base are the calling-bias control. (b) The *RNU4-2* / *RNU4-1* alignment over the critical region, showing the identical T_4 tract and the four paralogue differences all lying outside it. (c) Parental origin of informative ReNU cases in both published cohorts.

Data files

file	contents
results/renu_recurrence_rates.tsv	per-variant Roulette rates and observed carrier counts
results/homopolymer_insertion_rate.tsv	insertion frequency by tract base and length (short read)
results/homopolymer_rate_lr_vs_sr.tsv	long-read versus short-read rate at each tract length
results/gene_conversion_test.tsv	conversion-explicability of every observed patient allele
results/position_matched.tsv	within-position contrasts across the critical region
results/position_matched_sensitivity.tsv	residual as the substitution comparator widens
results/paralogue_control.tsv	mutational supply at the homologous <i>RNU4-1</i> tract
results/selection_requirement.tsv	per-division advantage a selective account would need

file	contents
data/processed/roulette_rnu4_2.tsv	all 423 Roulette records for <i>RNU4-2</i>
data/curation/redu_variant_counts.tsv	transcribed per-variant carrier counts

Methods

Mutation rate estimates. Base-pair-resolution germline mutation rates were obtained from Roulette (v5.2, TFBS-corrected) by HTTP range request against the per-chromosome bgzipped VCF using its CSI index, so the 3.8 GB file was not downloaded. Implementation in `snrna_vep/remote_tabix.py` [48].

Homopolymer insertion frequency. Ten windows of 200 kb were sampled uniformly at random from chr12:20–110 Mb (seed 0), excluding windows more than 20% N. Reference sequence was retrieved from the UCSC REST API and homopolymer tracts of length ≥ 2 enumerated by regular expression. gnomAD v4.1 genome sites were retrieved for each window by tabix range query against the 26 GB per-chromosome VCF. PASS variants were classified as substitutions (single-base reference and alternate) or single-base insertions (alternate one base longer and reference-prefixed), and each insertion was assigned to the homopolymer tract it extends, if any. Relative frequency is the number of tracts of a given base and length carrying an observed insertion, divided by the number of such tracts, expressed as a multiple of the number of observed substitutions divided by three times the number of callable bases [49]. The identical procedure, with the same seed and the same windows, was applied to the HPRC Release 2 (v2.1) Minigraph-Cactus pangenome [50].

Gene conversion test. Mature transcribed-strand sequences for *RNU4-2* and *RNU4-1* were taken from the GRCh38 reference, flanks trimmed and minus-strand genes reverse-complemented, then compared position by position. An observed patient substitution is conversion-explicable only if the donor carries the alternate base at the aligned position; an insertion is conversion-explicable only if the donor carries a longer homopolymer run at that site [56].

Paralogue control. gnomAD v4.1 PASS records were taken for both *RNU4-2* and *RNU4-1* over their gene bodies and over the homologous T_4 tract, whose coordinates follow from the shared minus-strand numbering (genomic = gene end – $n + 1$). Insertions were restricted to single-base events. Distinct observed alleles are the primary unit, matching the homopolymer analysis; allele counts are reported alongside because they are additionally shaped by drift and allele age [60].

Selection requirement. The per-division advantage implied by an R -fold enrichment over n divisions is $R^{1/n}$; reported for $n = 20, 30, 40$ and 60 oogonial divisions, and compared with the same quantity for a thousand-fold spermatogonial enrichment over approximately 610 divisions by paternal age forty [60].

Parental origin was taken from the published cohort reports and is not re-derived here.

Patient carrier counts were transcribed from the published Extended Data table of the discovery cohort and are provided in `data/curation/redu_variant_counts.tsv`.

Statistics. Spearman correlation for rank association; linear regression on $\log_{10}$ relative frequency against tract length; exact Poisson intervals on insertion counts.

Code and data availability. All analyses are scripted and committed; each result names its script.